
Cylindrical Flow Matching for Complex-Valued Medical Image Synthesis

Anonymous Author(s)

Affiliation

Address

email

Abstract

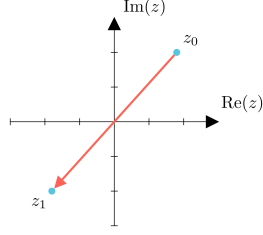
1 While deep generative models have achieved remarkable success in MRI recon-
2 struction, most operate strictly on magnitude images, discarding critical phase
3 information. This magnitude-only paradigm hinders the development of frequency-
4 domain Explainable AI (XAI) and complex-valued neural networks. Current
5 methods attempting to synthesize fully complex data suffer from a fundamental
6 topological mismatch: they naively embed the complex signal into a flat Euclidean
7 space ($\mathbb{R}^2$). This ignores the intrinsic circular topology of the phase (S^1), leading
8 to destructive interference, magnitude attenuation, and non-physical artifacts. To
9 address this, we propose **Cylindrical Flow Matching**, a geometry-aware gener-
10 ative framework that explicitly models amplitude on $\mathbb{R}^+$ and phase on S^1 . By
11 constructing probability paths along geodesics on a cylindrical manifold, our ap-
12 proach natively resolves phase discontinuities while preserving the computational
13 efficiency of Optimal Transport. We demonstrate that Cylindrical Flow Matching
14 ensures topological validity and physical signal plausibility, providing a reliable
15 synthetic baseline for safety-critical medical downstream tasks.

16 1 Introduction and Problem Formulation

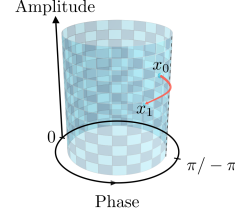
17 Magnetic Resonance Imaging (MRI) is a cornerstone of non-invasive medical diagnostics, yet its
18 inherently slow acquisition time remains a critical bottleneck. Accelerating MRI typically involves
19 undersampling the k -space and solving a highly ill-posed inverse problem [Chung and Ye, 2022].
20 While deep generative models have achieved remarkable success in MRI reconstruction, the vast
21 majority of these approaches operate strictly on magnitude images in the spatial domain, entirely
22 discarding the complex phase information.

23 This magnitude-only paradigm creates a significant research gap. Recently, emerging diagnostic
24 systems have begun to explore the immense potential of operating directly on complex-valued data
25 in the frequency domain, demonstrating that complex neural networks can better preserve signal
26 fidelity and crucial physiological information in raw MRI processing [Rempe et al., 2026, Dou et al.,
27 2025]. However, the lack of realistic, fully complex synthetic k -space data severely hinders the broad
28 development and validation of such architectures. Although recent diffusion-based approaches have
29 begun to explore complex-valued data generation to address this scarcity [Rempe et al., 2025], robust
30 and geometrically accurate generative baselines are still lacking. Furthermore, while state-of-the-art
31 counterfactual generators provide valuable interpretability in the spatial domain [Bedel and Çukur,
32 2024, Min et al., 2025], their lack of complex-domain constraints makes them prone to hallucinating
33 non-physical pixel values. Upon Inverse Fast Fourier Transform (IFFT), these spatial modifications
34 violate topological validity and leak into measured k -space subspaces, creating high-frequency
35 artifacts.

36 To safely develop frequency-domain Explainable AI (XAI) and reliable generative models, frame-
37 works must synthesize fully complex data while respecting physical laws. Yet, current approaches



(a) Standard Euclidean Flow ($\mathbb{R}^2$)



(b) Cylindrical Flow Matching (Ours)

Figure 1: Topological mismatch in complex-valued generation. (a) Shows the complex plane ($\text{Re}(z), \text{Im}(z)$), where standard models treat complex signals as flat Euclidean vectors. Linear interpolation between states z_0 and z_1 forces paths through the origin, causing destructive phase interference ($|z_{1/2}| \ll \frac{1}{2}(|z_0| + |z_1|)$) and non-physical signal cancellation. (b) Cylindrical Flow Matching ($S^1 \times \mathbb{R}^+$), where the probability path naturally wraps around the periodic boundary on S^1 , preserving signal magnitude.

suffer from a fundamental topological mismatch. They naively treat the complex signal as a flat, two-channel tensor (concatenating real and imaginary parts) embedded in a Euclidean space $\mathbb{C} \cong \mathbb{R}^2$ [Chung and Ye, 2022]. This assumption is fundamentally flawed as it completely ignores the intrinsic circular topology (S^1) of the signal’s phase. Standard Euclidean interpolation suffers from destructive interference due to the triangle inequality:

$$|z(t)| = |(1-t)z_0 + tz_1| \leq (1-t)|z_0| + t|z_1| \quad (1)$$

with strict inequality whenever phases differ ($\theta_0 \neq \theta_1$). Measured empirically on the SKM-TEA dataset at $t = 0.5$, the mean amplitude ratio $\frac{|z_{1/2}|}{\frac{1}{2}(|z_0| + |z_1|)}$ drops to 0.8417, demonstrating an average $\sim 16\%$ non-physical signal power loss in real MRI data. Furthermore, such flat interpolation fails to capture the continuous wrapping at the boundaries ($-\pi$ and π). In the context of safety-critical medical downstream tasks, such geometric violations are disqualifying.

To bridge the gap between generative modeling and the structured nature of physical signals, we propose **Cylindrical Flow Matching**, a geometry-aware generative framework designed specifically for complex-valued medical image synthesis. Building on flow matching [Lipman et al., 2023] and its manifold extension [Chen and Lipman, 2024], we explicitly model the amplitude on $\mathbb{R}^+$ and the phase on a circular manifold S^1 , naturally respecting the periodic topology of the MRI phase signal.

Crucially, despite operating on a non-Euclidean cylindrical manifold, our approach preserves mathematical elegance and computational efficiency. The probability paths on S^1 are constructed along geodesics using logarithmic and exponential maps, resolving the transport to the shortest angular distance and effortlessly crossing periodic boundaries without the need for heuristic penalty terms [Chen and Lipman, 2024]. Using comprehensive raw k -space datasets such as SKM-TEA, we demonstrate that substituting the standard Euclidean manifold with a cylindrical one natively resolves phase discontinuities, ensuring topological validity and physical signal plausibility, and providing a reliable source of synthetic complex-valued data for downstream clinical applications.

2 Methodology: Cylindrical Flow Matching

To construct a topology-aware generative model for complex-valued MRI data, we formulate continuous normalizing flows directly on the intrinsic manifold of the signal. A complex signal $z = Ae^{i\theta} \in \mathbb{C}$ naturally resides on a cylindrical manifold $\mathcal{M} = \mathbb{R}^+ \times S^1$, equipped with the product metric $g = dA^2 + d\theta^2$. This deliberate choice decouples amplitude and phase transport: geodesics interpolate each component independently along the shortest path, preventing the amplitude collapse inherent to the isometric polar embedding $g_{\text{polar}} = dA^2 + A^2 d\theta^2 \cong \mathbb{C} \setminus \{0\}$, whose geodesics are straight lines through the origin.

While this manifold possesses a vanishing Riemann curvature ($K = 0$), it exhibits a non-trivial first fundamental group ($\pi_1(\mathcal{M}) = \mathbb{Z}$). This non-trivial topology is the fundamental source of Euclidean phase collapse in standard diffusion models: interpolating phase in a flat space $\mathbb{R}^2$ ignores the 2π -periodicity and forces paths through the singularity at the origin, destroying amplitude and inducing phase discontinuities. A significant computational advantage of our geometric formulation on $\mathcal{M}$ is

that it is parallelizable; consequently, the exponential and logarithmic maps admit closed-form, $\mathcal{O}(1)$ analytic evaluations without the need for costly numerical ODE geodesic integrators.

2.1 Continuous Network Embedding

To circumvent artificial jump discontinuities at the branch cut $[-\pi, \pi)$, our network operates on a 3-channel continuous trigonometric embedding $x = (A, \cos \theta, \sin \theta) \in \mathbb{R}^3$ as input. The neural network then predicts a 2-channel velocity field $u_t = (v_A, v_\theta) \in T_{(A, \theta)}\mathcal{M}$, corresponding to the tangent space of the amplitude and phase.

2.2 Loss Formulation

Given paired samples $x_0 = (A_0, \theta_0) \sim p_0$ and $x_1 = (A_1, \theta_1) \sim q$, we construct geodesic probability paths along each component: $A_t = (1 - t)A_0 + tA_1$ and $\theta_t = \exp_{\theta_0}^{S^1}(t u_\theta)$. The target velocity field on the product manifold is:

$$u_A = A_1 - A_0, \quad u_\theta = \text{atan2}(\sin(\theta_1 - \theta_0), \cos(\theta_1 - \theta_0)) \quad (2)$$

where u_θ computes the shortest signed angular difference via the S^1 logarithmic map, correctly resolving transport across the $\pm\pi$ boundary.

The network $v_\phi(t, x)$ is trained to regress the target velocity field $u_t(x_t)$ via an L_1 objective, which is specifically chosen to provide robust edge preservation in MRI images. Matching our implementation, the exact loss is given by:

$$\mathcal{L}(\phi) = \mathbb{E}_{t, x_0, x_1} [|v_A - u_A| + \lambda A_1 |v_\theta - u_\theta| + \gamma \mathcal{L}_{\text{HF}}] \quad (3)$$

Crucially, the phase loss is modulated by the target amplitude A_1 (with weight λ). This amplitude mask is mean-normalized across each batch to stabilize training gradients, and gracefully prevents gradient explosion in noise or air background regions where the signal phase is undefined.

Furthermore, we apply a spatial-frequency regularization term to enhance sharp anatomical features: $\mathcal{L}_{\text{HF}} = \mathbb{E}[|\mathcal{F}(v_\phi - u_t)| \odot M(k)]$, where the velocity error is transformed into the frequency domain via a 2D Fast Fourier Transform and penalized by a radial mask $M(k)$ that linearly increases towards the periphery.

2.3 Base Distribution and Inference

We construct our generative process from the base distribution $p_0 = \mathcal{U}([0, 1]) \otimes \mathcal{U}(S^1)$, a uniform prior on the unit cylinder (clamped with $\epsilon = 10^{-6}$ for numerical stability at the origin). This ensures there is zero probability mass strictly at the origin singularity $A = 0$, effectively sidestepping topological breakdowns.

During inference, we simulate the learned vector field to generate samples. Given the analytic formulation of our manifold geometry, we employ a 2nd-order Heun predictor-corrector numerical integration scheme to solve the probability flow ODE accurately and efficiently.

3 Experiments and Results

We evaluate Cylindrical Flow Matching on complex-valued MRI synthesis and downstream reconstruction using the comprehensive raw k -space SKM-TEA dataset [Desai et al., 2021]. We compare our geometry-aware approach directly against the Euclidean baseline. Crucially, the training and inference pipelines, including the underlying neural vector field architecture, remain exactly the same for both methods; the singular modification is the underlying manifold topology.

As shown in Figure 2, the Euclidean model struggles with phase wrapping, leading to artificial signal cancellations and magnitude dropouts near the boundaries. In contrast, our geometry-aware approach accurately recovers the phase signal, preserving fine-grained high-frequency details essential for downstream medical diagnostics.

To properly evaluate phase reconstruction, we employ the amplitude-weighted Geodesic Phase Error (GPE), explicitly defined as:

$$\text{GPE}(\theta, \hat{\theta}) = \frac{\sum_i A_i \cdot (1 - \cos(\theta_i - \hat{\theta}_i))}{\sum_i A_i} \quad (4)$$



Figure 2: **Qualitative comparison.** From left to right: Real MRI phase, Euclidean Flow reconstruction (showing severe boundary artifacts), and our Cylindrical Flow reconstruction (preserving native topology).

Table 1: **Quantitative metrics.** Evaluation benchmark on generation and downstream reconstruction (ongoing runs denoted by –).

Method	Generation		Reconstruction	
	FID ↓	Geodesic Phase Error ↓	PSNR ↑	SSIM ↑
PhaseGen [Rempe et al., 2025]	–	–	–	–
Euclidean ($\mathbb{R}^2$)	–	–	–	–
Cylindrical Flow (Ours)	–	–	–	–

where A_i is the ground-truth magnitude. This formulation correctly accounts for the cyclic topology of S^1 . Naive Euclidean "Phase MSE" is invalid because it improperly penalizes topologically equivalent angles (e.g., $-\pi$ and π) with a maximum squared error of $(2\pi)^2$ instead of 0. Table 1 outlines our evaluation benchmark across both generation and reconstruction tasks (ongoing final runs denoted by –).

For the reconstruction task, we employ the trained cylindrical flow as a learned generative prior within a posterior sampling framework at $4\times$ Cartesian acceleration.

4 Conclusion and Future Work

By formulating probability paths on $\mathbb{R}^+ \times S^1$, Cylindrical Flow Matching natively resolves phase discontinuities and magnitude attenuation, providing a physically consistent generative baseline for complex-valued MRI. Future work will extend this framework to multi-coil acceleration benchmarks and frequency-domain counterfactual Explainable AI (XAI) on raw k -space [Bedel and Çukur, 2024, Min et al., 2025], bypassing spatial hallucination issues. Finally, extending this cylindrical manifold formulation to continuous Schrödinger Bridges will enable data-driven priors for downstream tasks such as complex medical image segmentation.

References

- Hasan Atakan Bedel and Tolga Çukur. DreaMR: Diffusion-driven counterfactual explanation for functional MRI. *IEEE Transactions on Medical Imaging*, 44(9):3654–3669, 2024. doi: 10.1109/TMI.2024.3475877.
- Ricky T. Q. Chen and Yaron Lipman. Flow matching on general geometries. In *International Conference on Learning Representations (ICLR)*, 2024.
- Hyungjin Chung and Jong Chul Ye. Score-based diffusion models for accelerated MRI. *Medical Image Analysis*, 80:102479, 2022. doi: 10.1016/j.media.2022.102479.
- Arjun D. Desai, Garry E. Gold, Brian A. Hargreaves, and Akshay S. Chaudhari. SKM-TEA: A dataset for accelerated MRI reconstruction with dense image labels for quantitative clinical evaluation. In *Advances in Neural Information Processing Systems (NeurIPS) Track on Datasets and Benchmarks*, volume 34, pages 25375–25389, 2021.
- Quan Dou, Zhixing Wang, Xue Feng, Adrienne E. Campbell-Washburn, John P. Mugler, III, and Craig H. Meyer. MRI denoising with a non-blind deep complex-valued convolutional neural network. *NMR in Biomedicine*, 38(1):e5291, 2025. doi: 10.1002/nbm.5291.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023.
- Hyungi Min, Taeseung You, Hangeul Lee, Yeongjae Cho, and Sungzoon Cho. InstructX2X: An interpretable local editing model for counterfactual medical image generation. In *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2025.
- Moritz Rempe, Fabian Hörst, Helmut Becker, Marco Schlimbach, Lukas T. Rotkopf, Kevin Kröninger, and Jens Kleesiek. PhaseGen: A diffusion-based approach for complex-valued MRI data generation. *arXiv preprint arXiv:2504.07560*, 2025.
- Moritz Rempe, Lukas T. Rotkopf, Marco Schlimbach, Helmut Becker, Fabian Hörst, Johannes Haubold, Philipp Dammann, Kevin Kröninger, and Jens Kleesiek. Efficient complex-valued vision transformers for MRI classification directly from k -space. *arXiv preprint arXiv:2601.18392*, 2026.